Technical audit — Nhill (baseline 0.1.0, target version 2.0.0)

> **Date:** 26 October 2026 · **Scope:** complete repository delivered (`nhill-0.1.0-src.tar.gz`) and the 0.1.0 deliverables (`nhill-release/`) · **Method:** full read of the tree, automatic measurements (lines, routes, tables, tests), execution of the tests that could run, execution of the delivered 0.1.0 binaries under Xvfb. **Major limit of the audit environment:** only GitHub is reachable; the Ubuntu/Debian, Qt, PyPI, npm and Google repositories are blocked, and the session security policy refused to build the dependencies from their sources. The complete project (server + Qt applications) **could therefore not be rebuilt here**. What was executed is stated explicitly.

1. Summary

Nhill is a multi-establishment business management suite written in **C++17**: an HTTP/JSON server and three **Qt 6 Quick (QML)** applications sharing one library. Overall quality is **good and above average** for a product at this stage: clean client/server split, exact money arithmetic in minor units, establishment isolation through forced PostgreSQL row-level security, versioned migrations with checksums, audit logging, real integration tests against a throwaway PostgreSQL.

The main weaknesses are not in the business logic but in:

1. **the delivery chain** (Windows and Android never built; Debian 13 packages built on Ubuntu with a frozen embedded Qt runtime that receives no security updates);
2. **presentation ergonomics** (fixed window sizes, truncated tables, insufficient contrast — see the UI/UX audit);
3. **vulnerable embedded dependencies** (cpp-httplib 0.14, frozen Qt 6.4.2 — see the security audit);
4. **no account recovery path** (no e-mail delivery).

Area	Assessment	Highlights
Architecture	Solid	Thin clients, authoritative server, one code base for 3 applications
Business logic	Solid, tested	Exact amounts, taxes, transactional numbering, issued documents immutable
Data	Solid	10 migrations, 64 tables, RLS forced on 30 business tables
Tests	Good	4 suites (common, server, setup, client); 0.1.0: 119 tests, 1 skipped
Build	Fragile	Only Linux built; reliance on scripts outside the repository (fixed in 2.0.0)
Distribution	Incomplete (0.1.0)	Windows/Android not built; Debian 13 with embedded runtime — fixed in 2.0.0: Windows installers, APK, native Debian 13 packages built in CI (validation report)
Observability	Minimal	`/health`, systemd journal, console “Technical errors”

2. Inventory

2.1 Languages, standards and size

Item	Measured value
Language	C++17 (`CMAKE_CXX_STANDARD 17`, extensions off)

Item	Measured value
Server	10,765 lines of C++ (`server/src`), **182 HTTP routes** registered
Client (C++)	1,500 lines (`client/src/core`, 3 `main.cpp`)
Client (QML)	8,955 lines, 95 files, 4 modules (`Nhill.Ui`, `Nhill.Business`, `Nhill.Console`, `Nhill.Pos`)
Shared library	`common/`: money, establishment codes, validation (≈ 690 lines)
Database	10 SQL migrations (1,709 lines), **64 tables**, dedicated `nhill` schema
Translations	`client/i18n/nhill_fr.ts` / `nhill_en.ts` (≈ 1,900 messages), `server/i18n/{fr,en}.json`

2.2 Frameworks and libraries

Component	Used by	Reference version (0.1.0)	Note
Qt 6 (Core, Gui, Network, Qml, Quick, QuickControls2, Svg, LinguistTools, Test)	Client	6.4.2 (Ubuntu 24.04)	"Basic" style, in-house `Fb*` components
QtKeychain (Qt 6)	Client	0.14 (optional)	Refresh tokens in the OS keychain; no clear-text fallback
libpqxx	Server	7.8	PostgreSQL access
PostgreSQL	Server	15+ (tests: 16; production: Supabase 17.6)	Forced RLS, dedicated roles
cpp-httplib	Server, client tests	0.14	**Vulnerable** (CVE-2025-46728, CVE-2025-53629), see security audit
nlohmann/json	Server	≥ 3.10	
libsodium	Server	1.0.18	Argon2id, secretbox, HMAC
OpenSSL	Server	3.x	Hashes, outbound TLS (payment providers)
libharu (libhpdf)	Server	2.3.0	Invoice, receipt and report PDFs
libpng	Server	1.6	Logos
GoogleTest	Tests	1.14	

2.3 Modules and entry points

```
common/            nhill_common (static)     money, taxes, conversions, codes, validation
server/src/        nhill_server_core + nhill-server
  main.cpp           sub-commands: serve, migrate, setup (installation mode)
  http/app.*         HTTP server, context, role permissions, JSON errors
  services/*.cpp     auth, establishments, members, subscriptions, providers, console,
                     support, POS, catalogue, parties, stock, sales, purchases, payments,
                     expenses, reports, activity, administration
  pdf/, jobs/, setup.cpp, migrate.cpp, util/crypto.cpp
client/src/        nhill_client_core (ApiClient, Session, Formatter, Validator, AppContext)
  business/ pos/ console/ main.cpp   three executables
client/qml/Nhill/  Ui (components), Business, Console, Pos (screens)
db/migrations/     versioned schema 0001–0010 (SHA-256 checksums verified)
tests/             common/, server/ (API + throwaway PostgreSQL, installation), client/
deploy/            systemd, Caddy (TLS), encrypted backup/restore
```

2.4 Build system

- **CMake ≥ 3.21 + Ninja**, options `NHILL_BUILD_SERVER/CLIENT/TESTS`, `NHILL_WARNINGS_AS_ERRORS`. Strict warnings (`-Wall -Wextra`, `/W4`).
- QML modules compiled by `qt_add_qml_module` (qmlcachegen); resources (IBM Plex fonts, SVG logos) embedded; translations compiled by `qt_add_lrelease`.
- Packages: CPack (`cmake/NhillPackaging.cmake`) and, in the 0.1.0 deliverables, `build-deb.sh` / `build-deb-debian13.sh` scripts **located outside the source repository** (fixed: moved into `packaging/` in 2.0.0).
- Windows: PowerShell script + NSIS/Inno Setup **never executed**; Android: `build-android.sh` **never executed**.

2.5 System dependencies (runtime)

- **Applications:** Qt 6 and its QML modules, xcb (X11) or Wayland platform plugin, OpenGL, fontconfig, D-Bus/Secret Service (keychain).
- **Server:** libpq5, libsodium23, libssl3, libhpdf, libpng, zlib, systemd, `nhill` system user.
- Check performed: the delivered 0.1.0 binaries start here with the embedded Qt runtime; this machine lacks `libOpenGL.so.0` and `libb2.so.1` (provided by `libopengl0` and `libb2-1` on a normal machine; replaced here by test shims).

2.6 Database

- PostgreSQL 15+; `nhill` schema closed to `PUBLIC`, `anon`, `authenticated`, `service_role`.
- Roles: `nhill_owner` (migrations), `nhill_app` (application, `NOBYPASSRLS`, owns no object), `nhill_backup` (backups).
- RLS forced on 30 business tables (`establishment_id = nh_current_establishment()`), composite foreign keys preventing cross-establishment references.
- 18 platform tables without RLS by design (sessions, subscriptions, logs…), protected by the server's queries (review documented in `docs/archive/securite.md`).
- Immutable migrations: a migration changed after being applied is refused.

2.7 Configuration

- Server: `/etc/nhill/server.env` (0600, refused if readable by others), `NH_*` variables (database, listening address, secrets, Argon2, token lifetimes, background jobs). Example without secrets: `config/server.env.example`.
- Clients: Qt preferences (`QSettings`) — server address, language, theme. **No business data is stored locally**, no database secret in the applications.
- Installation mode: a server without a database starts in `setup` mode and prints a one-time token in its log; Nhill Console drives the installation.

2.8 Authentication, authorisation, sessions

- Argon2id passwords (libsodium, 3 passes, 64 MiB), constant-time comparison, dummy hash for unknown e-mails.
- Opaque tokens (32 bytes) stored as SHA-256; access 15 min, refresh 30 days with rotation and reuse detection (family revocation).
- Establishment roles: `owner`, `admin`, `manager`, `sales`, `accountant`, `viewer`, `cash_manager`, `cashier`; permissions checked server-side (`App::roleHas`).
- Console: separate accounts, password + **mandatory TOTP**, roles `superadmin`, `billing`, `support`, `auditor`, token in memory only.

- Rate limiting (login, codes, invitations, console, POS…) stored in the database (`auth_throttle`), so it holds across processes.
- **Gap:** no “forgot password” flow (no e-mail transport).

2.9 Errors and logs

- Normalised JSON API errors (`code`, `message`, `details`, `request_id`); SQL errors mapped (RLS → 404 without leaking, constraints → stable codes).
- Logs on standard output (journald); “log transport” e-mails masked.
- The applications send their QML errors to the server (personal data masked); the console groups them by fingerprint.
- **Finding:** these reports revealed two real errors in 0.1.0 (TypeError in `PosSession.qml` and `total` of `null` in Nhill Business); the first one is fixed in 2.0.0.

2.10 Network, resources, online operation

- HTTP/JSON on `/api/v1`, TLS terminated by Caddy (HSTS, security headers, 6 MB body limit); the server listens on 127.0.0.1 only.
- **Online-only operation**: no offline mode (documented choice: stock, numbering and rights are arbitrated by the server). POS sales are protected against duplicates by an idempotency key. A strong constraint for the target market (see the market study).

3. Initial build and test status (baseline)

Check	Result	Status
Business tests `tests/common` (20 tests in 2.0.0, 18 in 0.1.0) built with g++ 13 and a GoogleTest-compatible runner (`scripts/test-common-standalone.sh`)	0.1.0: **18/18 passed**; 2.0.0: **20/20 passed**	Done and tested
QML static check (`scripts/qa/qml-check.py`: delimiter balance, JavaScript parsing of 4,201 bindings/functions by Node.js, modules, translations)	0.1.0: 0 errors; 2.0.0: 0 errors, 0 untranslated strings	Done and tested
Start of the delivered 0.1.0 binaries (Xvfb 1366×768)	Nhill POS, Business, Console start and show the sign-in screen	Done and tested
Full build (server + clients) and `ctest`	**Not executable here** (dependencies unavailable)	Blocked — run by the provided CI (`.github/workflows/ci.yml`)
Project test report 0.1.0	18 + 82 (1 skipped) + 6 + 13 passed	Taken from `docs/archive/rapport-tests.md`

4. Defects and technical debt found

#	Finding	Severity	2.0.0 treatment
T1	Packaging scripts of the 0.1.0 delivery outside the source repository	Medium	Moved into `packaging/`; versions read from `CMakeLists.txt`
T2	Server tests tied to `/usr/lib/postgresql/16` (fail on Debian 13, PostgreSQL 17)	Medium	Automatic version detection (`pgBinDir`)

#	Finding	Severity	2.0.0 treatment
T3	Windows and Android never built	High (distribution)	Reproducible CI workflows; to be run (see the build guide)
T4	Debian 13 packages with frozen embedded Qt 6.4.2/ICU/libjpeg	High (security)	Native Debian 13 build (system Qt 6.8) in CI; embedded variant discouraged
T5	Vulnerable cpp-httplib 0.14 (DoS through chunked bodies)	High	Mitigated by Caddy (`request_body max_size`); upgrade to ≥ 0.23 recommended (not done: cannot be tested here)
T6	Fixed window sizes (1280×800/820, 1320×840)	High (UX)	Screen-fitted size, centring, maximised on small screens
T7	`X-Forwarded-For` read from the left (forgeable behind an appending proxy)	Low (the provided Caddy replaces the header)	Right-most entry used
T8	POS, support and console actions without labels in the activity log	Medium (UX/ traceability)	51 FR/EN labels added; technical identifiers hidden
T9	No account recovery path	High (operations)	Not fixed (needs e-mail delivery) — design proposed, to be planned
T10	No offline mode	Medium (market)	Out of 2.0.0 scope (architecture change)
T11	No DGI standardised invoicing (SFE/ DEF)	High (DRC compliance)	Out of scope; homologation roadmap
T12	Duplicated `scripts/build-*.ps1/ sh` and deliverables	Low	Single entry point in `scripts/`, implementation in `packaging/`
T13	QML fields whose width depends on the typed text (unbalanced grids)	Medium (UX)	Stable preferred width for fields
T14	Race in the POS ticket dialog (`full` null)	Medium	Guard added
T15	No QML checking tool without Qt	Low	`scripts/qa/qml-check.py`

5. 2.0.0 changes that may affect business behaviour

In line with the rule of preserving business logic, every change is listed with its justification and its test:

Change	Justification	Business effect	Test
Minimum password length 10 → **8** (14 → 8 for the console)	Requirement of the product owner	Authentication rule relaxed; refusal of common, repetitive or e-mail-equal passwords **kept**; console TOTP **kept**	`Validation.PasswordMinimumIsEightCharacters`, `Validation.Auth.PasswordMinimumIsEightCharactersOnEveryPath`, `Admin.`
Extended list of common passwords (8–9 characters)	Compensates the shorter length	A few weak passwords refused	Tests above

Change	Justification	Business effect	Test
`X-Forwarded-For`: right-most entry	Security fix	Reliable per-IP limits whatever the proxy	Logic tested in isolation (6 cases)
Activity log summary without technical identifiers	Readability, minimisation	None (raw data unchanged in the database and the API)	Code review
Cash count check before closing a POS session	The server already requires it (`cash_count_required`)	None (earlier message)	Code review

No change to calculations, data model, migrations, invoicing and subscription rules, permissions or API formats.

6. Prioritised recommendations

1. **Run the provided CI** (build, 4 test suites, native Ubuntu 24.04 and Debian 13 packages, Windows, Android) and publish the artefacts — the only step that turns the 2.0.0 code into verified deliverables.
2. Upgrade cpp-httplib (≥ 0.23, single header vendored in the repository) and drop the embedded Qt runtime for Debian 13.
3. Implement e-mail delivery (SMTP) then password reset (single-use, short-lived link, 8-character rule).
4. Start the SFE/DEF homologation with the DGI before targeting VAT-registered businesses.
5. Study a degraded offline mode for Nhill POS (encrypted local queue, provisional numbering) — an architecture change to design carefully.
6. Add monitoring (metrics, backup alerts) before going to production.